TEXORA · LECTURE SHEET

Core Yarn

A core-spun yarn hides a continuous filament at its centre and wraps it in staple fibre, so the fabric keeps the look and feel of cotton or wool while the invisible core supplies stretch or strength. This sheet covers the structure, ring-frame production with elastane pre-draft, core-content calculation, properties and uses.

YM-4103 · SPECIAL YARN MANUFACTURING · TOPIC 5 · LEVEL 4 TERM 1 · YARN MANUFACTURING SPECIALIZATION

1. Principle and structure

A **core yarn** (core-spun yarn) consists of a continuous central filament — the **core** — completely covered by a sheath of staple fibres spun around it in the same operation. The two components divide the work exactly as in the fancy-yarn model of Topic 2, but here the aim is the opposite of decoration: the core must be *invisible*. The sheath gives the yarn its textile surface — appearance, absorbency, dyeing behaviour and hand — while the core gives a bulk property the staple fibre cannot supply:

- **Elastane (spandex) core** — stretch and elastic recovery: stretch denim, stretch shirting, hosiery, waistbands, activewear.
- **High-tenacity filament core** (polyester or nylon) — strength: the classic core-spun sewing thread, where a polyester filament core wrapped in cotton or polyester staple combines filament strength with a fibre surface that protects the thread in the needle and bonds well in the seam.
- **Other functional cores** — conductive, cut-resistant or soluble filaments for technical yarns.

Quality is defined by **coverage**: the core must sit centrally and remain covered along the whole length. A core displaced to the surface (“grinning” or bare patches) shows as a shiny filament streak, dyes differently, and in elastane yarns destroys uniform stretch.

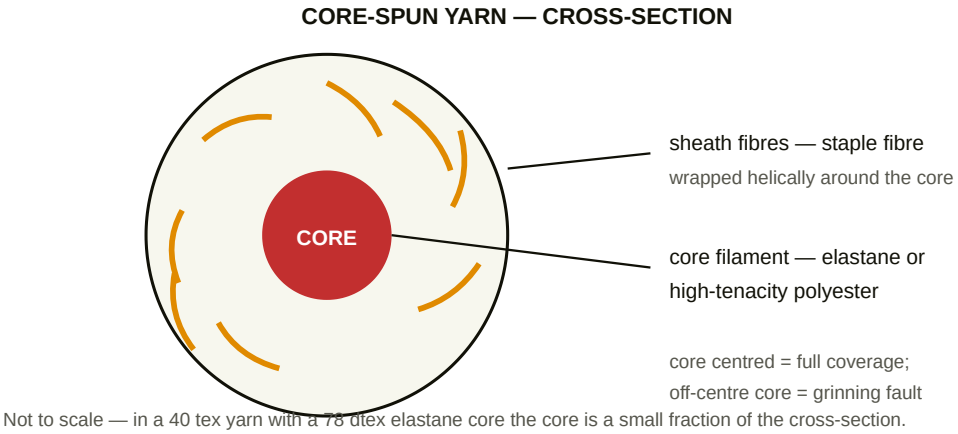


Figure 1 — Cross-section of a core-spun yarn. A single central filament is completely surrounded by staple sheath fibres; properties come from the core, surface character from the sheath.

2. Production on a modified ring frame

Core-spun yarn is produced on a ring frame carrying two additions: a core feed system and, for elastane, a controlled pre-draft unit.

1. **Sheath preparation.** Staple roving is drafted normally through the ring-frame drafting system to the required sheath fineness.
2. **Core feed.** The filament is unwound from its package over a guide and fed by driven feed rollers positioned so the core enters the drafted strand exactly at the centre of the front-roller nip. Centring is critical: the core guide must align with the middle of the fibre ribbon, and a V-grooved front roller or guide is often used to hold that position.
3. **Elastane pre-draft.** Elastane is never fed relaxed. Positive feed rollers stretch it by a set **pre-draft** (draft ratio, commonly around 3–4× for apparel stretch yarns) so that in the finished yarn it is held extended; when the fabric is released, the core tries to retract and provides stretch and recovery. The pre-draft also thins the core: its linear density inside the yarn is the package linear density divided by the pre-draft (worked example below).
4. **Twist insertion and coverage.** As the combined strand leaves the front rollers, spindle twist wraps the sheath fibres helically around the core. Twist level must be sufficient to lock coverage under fabric strain — an elastane fabric stretched in wear must still not expose the core — but not so high that the yarn becomes hard and the stretch is strangled.
5. **Winding and clearing.** The yarn is wound and electronically cleared; bare-core and off-centre faults that clearing cannot detect are controlled by feed alignment and monitored by periodic cross-section/coverage checks.

Alternative routes — hollow-spindle wrapping of a sheath yarn around a core, and friction (DREF) core spinning for coarse technical yarns — follow the same principle with different sheath-locking

mechanisms, but modified ring spinning dominates apparel core yarns.

WORKED EXAMPLE — CORE CONTENT OF AN ELASTANE CORE-SPUN YARN

Finished yarn: 40 tex core-spun. Core: elastane of 78 dtex, pre-drafted 3.5×.

Step 1 — convert core to tex: 78 dtex = 78 / 10 = **7.8 tex** (as supplied).

Step 2 — core tex inside the yarn: pre-draft stretches each length 3.5×, so the same mass occupies 3.5× the length: 7.8 / 3.5 = **2.23 tex**.

Step 3 — core content = 2.23 / 40 × 100 = **5.6%** elastane; the staple sheath is the remaining 94.4% (37.77 tex).

Design reading: only about one-eighteenth of the yarn mass is elastane, yet that small core controls the stretch of the whole fabric — which is why core % and pre-draft, not sheath count, are the stretch-design variables. Raising pre-draft at fixed package size *reduces* core content while increasing retractive force per unit of core.

3. Properties

- **Stretch and recovery (elastane core)** — fabric stretch is governed by core %, pre-draft, yarn count and fabric structure together; recovery (return after extension) is the elastane's key gift, but it decays with heat, chlorine and age, and sheath slippage under repeated stretch shows as growth or bagging if coverage/twist are inadequate.
- **Strength (filament core)** — tenacity approaches that of the filament core while the surface remains staple-like; sewing threads exploit exactly this split.
- **Appearance and handle** — those of the sheath fibre, provided coverage is complete: cotton-sheath stretch denim looks, feels and dyes as cotton.
- **Processing notes** — elastane yarns must be wound and handled under controlled tension (a relaxed package lets the core retract and snarls the sheath); fabrics need heat-setting control so stretch is stabilised without degrading the elastane.

CORE	SHEATH	PROPERTY GAINED	MAIN PRODUCTS
Elastane 40–80 dtex, pre-draft ≈3–4×	Cotton	Comfort stretch, recovery	Stretch denim, shirting
Elastane	Polyester / textured blends	High stretch, easy care	Activewear, hosiery
High-tenacity polyester filament	Cotton or polyester staple	Tenacity + sewability	Core-spun sewing thread

Metal / conductive filament	Various staple	Conductivity with textile handle	Smart textiles, antistatic fabrics

4. Faults and their causes

- **Bare core / grinning** — core off-centre at the front roller, sheath supply too low, or twist too low; correct the core guide alignment first.
- **Sheath slippage (core pulls out)** — insufficient twist or too smooth a core filament; raise twist within handle limits.
- **Uneven stretch** — pre-draft variation from feed-roller slip or package tension variation; the fault appears in fabric as wavy stretch bands, not as a yarn-count fault.
- **Core breakage** — elastane over-drafted or damaged at guides; a broken core retracts and leaves a length of ordinary yarn — invisible until the fabric is stretched.

Glossary

Core-spun yarn — yarn with a continuous filament core fully covered by a staple sheath. **Pre-draft** — controlled stretching of elastane before it enters the yarn; package tex / pre-draft = core tex in yarn. **Core content %** — core tex in yarn / yarn tex × 100. **Coverage** — completeness with which the sheath hides the core. **Grinning** — core showing through the sheath. **Recovery** — return of a stretched yarn/fabric towards its original length. **Sheath slippage** — staple sheath sliding along the core under strain.

Key points

• Core gives the bulk property (stretch or strength); sheath gives the surface (look, feel, dyeing) — specify both.
• On the ring frame the core must enter dead-centre at the front-roller nip; alignment is the coverage control.
• Elastane is fed pre-drafted (commonly ≈3–4×): core tex in yarn = package tex / pre-draft.
• 40 tex yarn + 78 dtex core at 3.5× pre-draft → 2.23 tex core → 5.6% core content.
• Signature faults: grinning, sheath slippage, uneven stretch, hidden core breaks — all feed/twist faults, not count faults.

Practice

1. A 30 tex core-spun yarn uses a 70 dtex polyester-elastane core pre-drafted 3.0×. Calculate the core tex in the yarn and the core content %.

2. Stretch denim shows shiny filament streaks after weaving. Name the fault, give its most likely machine cause, and state the first adjustment to make.
3. Explain why doubling the elastane package size (dtex) at unchanged pre-draft and yarn count would not simply “double the stretch”, referring to core content and sheath behaviour.
4. State, with reasons, which component you would change to improve (a) seam strength of a sewing thread and (b) its surface behaviour in high-speed sewing.

Answer key

1. 70 dtex = 7.0 tex supplied. In yarn: 7.0 / 3.0 = **2.33 tex**. Core content = 2.33 / 30 × 100 = **7.8%**.
2. **Grinning / bare core**. Most likely cause: core entering **off-centre** at the front-roller nip (misaligned core guide), so the sheath wraps unevenly. First adjustment: re-align the core guide / V-groove so the core meets the centre of the drafted ribbon; check sheath supply and twist only after alignment is confirmed.
3. Core content would roughly double, but stretch is a system property: the same 30 tex sheath must now cover a thicker core, coverage and twist locking are strained, the yarn becomes harsher, and fabric stretch is limited by how far the woven structure and sheath can extend — beyond the design point, extra core mainly adds retractive tightness, boardiness and grinning risk rather than usable extra stretch. Stretch design adjusts core %, pre-draft, count and fabric structure together.
4. (a) The **core** — use a higher-tenacity (or larger) filament core, since seam strength follows the continuous filament. (b) The **sheath** — its fibre, fineness and coverage govern needle heating, friction and seam appearance; the core should stay hidden and unchanged for this purpose.
